

AI Smart Navigation Assistant for Visually Impaired People Using Raspberry Pi

Smart Cane Environmental Awareness System

Final Presentation · IoT Term Project

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Motivation & Problem

Visually impaired users face daily challenges navigating indoor spaces safely and independently — especially unfamiliar environments.

Existing solutions fall into two extremes:

Full SLAM systems

Accurate, but expensive: LiDAR, IMU, heavy compute. Out of reach for a low-cost student project.

Simple ultrasonic canes

Cheap, but "dumb": beep when close, no semantic information — no idea *what* or *where*.

“Our goal: meaningful semantic awareness, on a single Raspberry Pi, fully offline, at low cost.”

Project Idea

A smart cane that answers three questions in real time:

Question	How
What is in front of me?	ToF depth detection + YOLOv8-nano classification
Where am I?	ORB visual place recognition
How do I get to my destination?	Voice input + turn-by-turn audio guidance

"Where do you want to go?" → "Kitchen"
"Chair ahead, 0.8 metres." → "Turn left." → "You have arrived."

Scope Decision — From SLAM to Awareness

Original vision

Full Visual SLAM (ORB-SLAM3): build a 3D map in real time, plan routes autonomously.

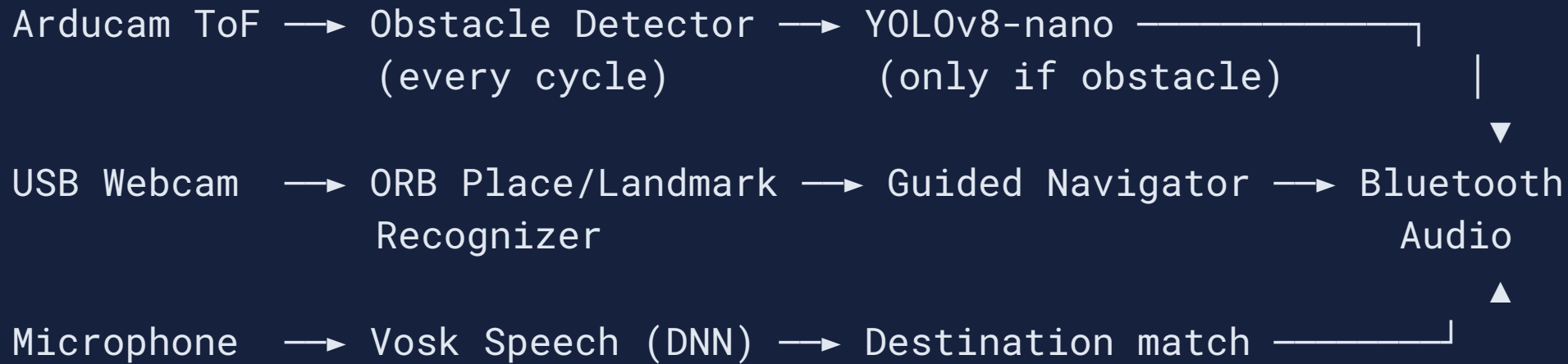
➡ Needs IMU, heavy compute, careful calibration — not achievable reliably in the available time.

Revised, shipped scope

Two achievable modes:

- ● **Awareness** — passive obstacle + location alerts
 - ● **Guided navigation** — voice destination + reactive turn-by-turn
- SLAM preserved as documented future work.

System Architecture



Stage	Technique	Runs
Obstacle detection	Depth threshold	Every cycle (2 Hz)
Obstacle classification	YOLOv8-nano (DL)	Only when flagged
Place/landmark recognition	ORB (classical CV)	Every 3rd cycle
Speech-to-text	Vosk (DL)	Once, at start

Mode 1 — Awareness

Passive loop: no destination, just continuous environmental feedback.

```
def _check_obstacles(self):
    alert = self._obstacle_detector.check(tof_frame)
    if alert:
        message = self._obstacle_detector.alert_message(alert)
        if self._object_classifier.is_available:
            detection = self._object_classifier.classify(webcam_frame)
            if detection:
                message = f"{detection.label} ahead, {alert.min_depth:.1f}m."
        self._audio.alert(message)
```

Output: *"You are in the office." · "Chair ahead, 0.8 metres."*

Mode 2 — Guided Navigation

No IMU/odometry → **reactive wall-following**, not path planning to coordinates.

```
ToF frame → split into LEFT / CENTER / RIGHT zones
center clear           → "Go straight"
center blocked, L > R  → "Turn left"
center blocked, R > L  → "Turn right"
both sides blocked     → "Path blocked. Please stop."
```

Combined with: **arrival detection** (ORB match vs. destination) and **landmark announcements** (trained doors, recognized + close ahead).

Works reliably for a known, previously walked route.

What If the User Deviates From the Route?

No map, no odometry → the system has **no concept of "on" or "off" route**, only reacts to what it currently sees.

Behavior	Result
Obstacle avoidance	✓ Keeps working — purely reactive to live ToF
Arrival / landmark detection	⚠ Can silently fail if the viewpoint differs too much from training
Recovery / "you seem lost"	✗ Not implemented — gives wall-following forever, no replanning

“This is exactly why lighting and viewpoint robustness matters — measured next.”

Hardware — Assembled Prototype



Hardware Components



Raspberry Pi 4
Central compute



Arducam ToF B0410
Depth, up to 4m



Logitech C270
Recognition + YOLO



BT Headphones
Audio I/O

Total hardware cost: ~**€80** — no GPU, no cloud, no internet required.

Software Architecture

Module	Responsibility
<code>sensors/</code>	Webcam + ToF abstraction, auto V4L2 device detection
<code>mapping/</code>	Waypoint store (SQLite + <code>.npy</code>); <code>location</code> vs <code>landmark</code>
<code>localization/</code>	ORB place/landmark recognition
<code>obstacle/</code>	ToF depth threshold
<code>perception/</code>	YOLOv8-nano obstacle classification (DL)
<code>speech/</code>	Vosk offline speech-to-text (DL)
<code>navigation/</code>	Reactive wall-following guidance
<code>audio/</code>	espeak-ng TTS + alert tones

27 automated unit tests across the codebase.

AI / Deep Learning Components

Technique	Type	Training data	Used for
ORB + BFMatcher	Classical CV	2-min capture, no labels	Place / landmark recognition
YOLOv8-nano	CNN (DL) , pretrained	None (COCO weights)	Classify flagged obstacles
Vosk	DNN-HMM (DL) , pretrained	None (public model)	Offline speech-to-text

Why gated, not continuous? Running a CNN every cycle is too slow for real-time RPi4 CPU. YOLO only runs *after* the cheap ToF threshold already flagged something close — classifying what's already known to be near.

Why not YOLO for doors? "Door" isn't a COCO class — ORB landmarks fill that gap with zero labeled data.

Dataset Sources & Accuracy

Technique	Dataset source	Published accuracy
ORB	None — self-collected from the user's own space (2-min capture)	N/A (not a learned model)
YOLOv8-nano	COCO — 80 classes, ~330K images, ~1.5M instances	mAP ₅₀₋₉₅ = 37.3 (COCO val2017, Ultralytics)
Vosk	Public English speech corpora (Alphacephei)	WER 9.85% (LibriSpeech test-clean)

Our own validation: since the Pi was unavailable for re-testing, we ran a controlled ORB robustness test using the *exact production matching code* on synthetic image perturbations (next slide).

ORB Robustness — Real Test Results

Same algorithm as production (`place_recognizer.py`): 500 ORB features, Lowe's ratio 0.75, `min_good_matches=15` .

Perturbation	Outcome
Rotation 0°–90°	Never failed — stayed above 375/500 good matches
Gaussian blur (motion)	Failed at kernel ≥ 15 (144 $\rightarrow$ 79 good matches)
Darker lighting	Failed at -40 brightness and below (133 $\rightarrow$ 27 matches)
Brighter lighting	Never failed — stayed ≥ 400 matches

“Rotation alone underestimates real viewpoint change (no perspective distortion modeled). **Blur and darkening are the more realistic failure modes** — directly explains the deviation risk from the previous slide.

”

Code — Gated DL Inference

perception/object_detector.py

```
def classify(self, bgr_frame: np.ndarray) -> Optional[DetectionResult]:
    if not self.is_available:
        return None

    results = self._model.predict(bgr_frame, verbose=False)[0]
    if results.bboxes is None or len(results.bboxes) == 0:
        return None

    best_idx = int(results.bboxes.conf.argmax())
    confidence = float(results.bboxes.conf[best_idx])
    if confidence < self._confidence_threshold:
        return None

    class_id = int(results.bboxes.cls[best_idx])
    return DetectionResult(label=self._model.names[class_id], confidence=confidence)
```

Code — Wall-Following Decision

navigation/guided_navigator.py

```
def _wall_following(self, left, center, right):
    if center <= 0 or center > self._clear_distance:
        return GuidanceResult(NavCommand.STRAIGHT, "Go straight.")
    if left <= 0 and right <= 0:
        return GuidanceResult(NavCommand.STOP, "Path blocked. Please stop.")
    if left > right:
        return GuidanceResult(NavCommand.TURN_LEFT, "Turn left.")
    elif right > left:
        return GuidanceResult(NavCommand.TURN_RIGHT, "Turn right.")
    else:
        return GuidanceResult(NavCommand.STOP, "Path blocked. Please stop.")
```

Pure decision logic — directly unit-testable without hardware.

Results & Current Status

Phase	Status
Sensor integration	✓ Complete — validated on real RPi hardware
Location training	✓ Complete — 1000–5000 ORB descriptors per location
Awareness loop	✓ Complete — running end-to-end
Guided navigation	✓ Implemented — voice, wall-following, landmarks, arrival
YOLO obstacle classification	✓ Implemented — gated DL inference

27/27 automated tests passing.

Contributions

- A working, **fully offline** assistive prototype combining depth sensing, classical CV, and **two deep learning models** on a single Raspberry Pi 4
- A deliberate architecture pairing **ORB + YOLO + Vosk**, each used where it performs best — not defaulting to one technique
- Robust hardware layer: **auto-detects camera devices by name**, solving real USB index instability found during development
- **27 automated unit tests** covering data model, sensor processing, navigation logic, and graceful DL degradation

Challenges & Solutions

- 1. Full SLAM was infeasible** — no IMU, no odometry, CPU-only Pi. We treated this as a scope decision, not a failure: redesigned around reactive wall-following + ORB recognition, kept SLAM as documented future work.
- 2. Training data evolved through 3 iterations** — single frame (unreliable) → 5-frame burst (still too similar) → 2-minute continuous capture, ~40 frames, up to 5000 descriptors (final).
- 3. Cross-platform threading bug, found *this week*** — building a desktop demo without the Pi revealed `pyttvx3` hangs when one engine is shared across threads on Windows (COM apartment threading). Fixed with a fresh engine per call.
- 4. DL inference too slow for continuous use** — gated YOLO behind the cheap ToF threshold, paying CNN cost only when an obstacle is already confirmed.

Future Work

Deliberately deferred from the original vision:

-  **Visual SLAM** — real-time 3D mapping (ORB-SLAM3), enabling navigation to unseen destinations
-  **Graph-based route planning** — `WaypointEdge` data model already exists; add Dijkstra across multiple legs
-  **IMU integration** — dead-reckoning between locations
-  **Fine-tuned object detection** — custom classes (doors, stairs, curbs) instead of relying on COCO + ORB
-  **Multi-environment support** — automatic environment selection

Thank You

Smart Cane Environmental Awareness System

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github.com/w1ggor/smart-nav-cane

Questions?